Energies of dilute Fermi gases

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Many-body quantum system with N particles in large box $\Lambda = [0, L]^3$ and Hamiltonian

$$H_N = \sum_{j=1}^{N} -\Delta_{x_j} + \sum_{1 \le i < j \le N} v(x_i - x_j).$$

- Kinetic energy: Laplacian $-\Delta_{x_j}$
- Interaction energy: Pairwise interaction $v \geq 0$, radial and of compact support.

At zero temperature / For ground state energy:

- Spin-$\frac{1}{2}$ Fermi gas: Realize H_n on $L^2_a(\Lambda^n; \mathbb{C}^2) = \bigwedge^n L^2(\Lambda; \mathbb{C}^2)$ or $L^2_a(\Lambda^{n\uparrow}) \otimes L^2_a(\Lambda^{n\downarrow})$ if fixed number of spin $\uparrow$ and $\downarrow$.
- Spin-polarized Fermi gas: Realize H_n on $L^2_a(\Lambda^n; \mathbb{C}) = \bigwedge^n L^2(\Lambda; \mathbb{C})$. (Can be realized experimentally using Feshbach resonances (for e.g. ^{6}Li, ^{40}K): "p-wave fermions")

At positive temperature / For the pressure: Realize on Fock spaces

$$\mathcal{F} = \bigoplus_{n=0}^{\infty} \bigwedge^n L^2(\Lambda; \mathbb{C}^2) \quad = \bigoplus_{n=0}^{\infty} L^2_a(\Lambda^n; \mathbb{C}^2) \quad \text{spin-1/2 fermions}$$

$$\mathcal{F} = \bigoplus_{n=0}^{\infty} \bigwedge^n L^2(\Lambda; \mathbb{C}) \quad = \bigoplus_{n=0}^{\infty} L^2_a(\Lambda^n; \mathbb{C}) \quad \text{spin-polarized fermions}$$

We study **ground state energy** in the thermodynamic limit

$$e(\rho_\uparrow, \rho_\downarrow) = \lim_{\substack{L\to\infty \\ N_\sigma/|\Lambda|\to\rho_\sigma}} \inf_{\substack{\psi\in L^2_a(\Lambda^{N_\uparrow})\otimes L^2_a(\Lambda^{N_\downarrow}) \\ \|\psi\|_{L^2}=1}} \frac{\langle\psi|H_N|\psi\rangle}{|\Lambda|}$$

$$e(\rho) = \lim_{\substack{L\to\infty \\ N/|\Lambda|\to\rho}} \inf_{\substack{\psi\in L^2_a(\Lambda^N) \\ \|\psi\|_{L^2}=1}} \frac{\langle\psi|H_N|\psi\rangle}{|\Lambda|},$$

or **pressure** in the thermodynamic limit

$$\psi(\beta,\mu) = \lim_{L\to\infty} \frac{1}{\beta|\Lambda|} \log \mathrm{Tr}_{\mathcal{F}} \left[e^{-\beta(\mathcal{H}-\mu\mathcal{N})}\right] = \lim_{L\to\infty} \sup_{\Gamma} P_\Lambda[\Gamma]$$

at temperature $T = 1/\beta > 0$ and chemical potential μ.

- $\mathcal{H} = \bigoplus_{n=0}^\infty H_n$ is the Hamiltonian on Fock space.
- $\mathcal{N} = \bigoplus_{n=0}^\infty n$ is the number operator.
- $P_\Lambda[\Gamma]$ is the **pressure functional** of the state Γ. A state Γ is an operator on $\mathcal{F}$ with $0 \le \Gamma \le 1$ and $\mathrm{Tr}_{\mathcal{F}}\,\Gamma = 1$.

$$L^3 P_\Lambda[\Gamma] = \mathrm{Tr}_{\mathcal{F}}[\mathcal{H}\Gamma] - \frac{1}{\beta} S(\Gamma) = \mathrm{Tr}_{\mathcal{F}}[\mathcal{H}\Gamma] + \frac{1}{\beta}\,\mathrm{Tr}_{\mathcal{F}}\left[\Gamma \log \Gamma\right].$$

Scattering lengths

Dilute limit: Interparticle spacing $\rho^{-1/3}$ large compared to lengthscale of interaction.

(Think, V has finite range R_0, then $\rho^{-1/3} \gg R_0$, i.e.; $\rho R_0^3 \ll 1$.)

- $\rightsquigarrow$ Expect only pairwise interactions (to leading order)
- $\rightsquigarrow$ Study minimal energy of two particles in a box with
  - no symmetry (fermions of different spin) $\rightsquigarrow$ s-wave: $E \simeq \frac{8\pi a_s}{L^3}$
  - fermionic symmetry (fermions of same spin) $\rightsquigarrow$ p-wave: $E \simeq \frac{4\pi^2}{L^2} + \frac{16\pi^3 a_p^3}{L^5}$

Definition

The s- and p-wave scattering lengths a_s and a_p are defined by

$$4\pi a_s = \inf\left\{\int \left(|\nabla f|^2 + \frac{1}{2} v f^2\right) \mathrm{d}x : f(x) \to 1 \text{ for } |x| \to \infty\right\},$$

$$12\pi a_p^3 = \inf\left\{\int \left(|\nabla f|^2 + \frac{1}{2} v f^2\right) |x|^2 \,\mathrm{d}x : f(x) \to 1 \text{ for } |x| \to \infty\right\}.$$

$$8\pi a_s = \int f_s v \approx \int v, \qquad 24\pi a_p^3 = \int |x|^2 f_p v \approx \int |x|^2 v \qquad \text{for smooth and small } v.$$

Two particles in a (periodic) box

Non-interacting ground state: (Easy exercise)

$$\psi_0(x,y) = \begin{cases} \dfrac{1}{L^3} & \text{for different spins (or bosons),} \\ \dfrac{2}{L^3}\cos\left(\dfrac{2\pi}{L}\,\dfrac{x^1+y^1}{2}\right)\sin\left(\dfrac{\pi}{L}(x^1-y^1)\right) & \text{for same spins,} \end{cases}$$

Interacting ground state is of form $\psi(x,y) = f(x-y)\psi_0(x,y)$ for some f. (By translation invariance.)

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Large volume limit $L \to \infty$: f minimizes (easy calculation)

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$\rightsquigarrow$ f is the s- or p-wave scattering function.

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$\rightsquigarrow$ f is the s- or p-wave scattering function.

The interacting ground state is given by the non-interacting ground state multiplied by the scattering function $f_{s/p}$.

Bosons

Realize $H_N = \sum_{j=1}^N -\Delta_{x_j} + \sum_{1\le i<j\le N} v(x_i - x_j)$ on $L^2_s(\Lambda^N) = L^2(\Lambda)^{\otimes_{\mathrm{sym}} N}$.

Theorem (Dyson 1957; Lieb–Yngvason 1998)

Let $v \ge 0$ be radial and of compact support. Then for $a_s^3\rho \ll 1$

$$e_{\mathrm{Bose}}(\rho) = 4\pi a_s \rho^2 \left[1 + o(1)\right], \qquad a_s^3\rho \ll 1.$$

- Energy of one pair of particles is $8\pi a_s/L^3$. (Short calculation)
- There are $N(N-1)/2 \simeq N^2/2$ many pairs.

$\rightsquigarrow$ Energy density $4\pi a_s \rho^2$.

Goal for fermions:

Find "analogue of $4\pi a_s \rho^2$" for both zero and positive temperature.

Spin-1/2 Fermi gas

Theorem (Lieb–Seiringer–Solovej 2005)

Let $v \geq 0$ be radial and of compact support. Then for $a_s^3 \rho \ll 1$

$$e(\rho_\uparrow, \rho_\downarrow) = \frac{3}{5}(6\pi^2)^{2/3}\left(\rho_\uparrow^{5/3} + \rho_\downarrow^{5/3}\right) + 8\pi a_s \rho_\uparrow \rho_\downarrow + o(a_s(\rho_\uparrow + \rho_\downarrow)^2).$$

Theorem (Seiringer 2006)

Let $v \geq 0$ be radial and of compact support. Then for small $a_s^3 \rho$ uniformly in $T = 1/\beta \lesssim \rho^{2/3}$

$$\psi(\beta, \mu) = \psi_0(\beta, \mu) - 2\pi a_s \rho^2 [1 + o(1)].$$

- Kinetic energy/pressure of free gas.
- Interaction energy. Similar to $4\pi a_s \rho^2$ for the Bose gas:
  - Energy of one pair of spin-$\uparrow$ and spin-$\downarrow$ is $8\pi a_s / L^3$. (Short calculation)
  - $N_\uparrow N_\downarrow$ many such pairs $\rightsquigarrow$ Energy density $8\pi a_s \rho_\uparrow \rho_\downarrow$.
- $T_F \sim \rho^{2/3}$ is the **Fermi temperature**. For higher temperatures expect thermal effects stronger than quantum and so gas should behave like classical gas.

Spin-polarized Fermi gas

Theorem (L.–Seiringer 2024)

Let $v \geq 0$ be radial and of compact support. Then for $a_p^3 \rho \ll 1$

$$e(\rho) \leq \frac{3}{5}(6\pi^2)^{2/3}\rho^{5/3} + \frac{12\pi}{5}(6\pi)^{2/3} a_p^3 \rho^{8/3}\left[1 - \frac{1}{14}(6\pi^2)^{2/3} a_p^3 R_{\text{eff}}^{-1} \rho^{2/3}(1 + o(1))\right].$$

$$R_{\text{eff}}^{-1} = \frac{1}{10\pi a_p^6}\int_{\mathbb{R}^3} |x|^4 \left(|\nabla f_p(x)|^2 + \frac{1}{2} v(x)|f_p(x)|^2\right) \mathrm{d}x \qquad \text{the effective range.}$$

Theorem (L.–Seiringer 2024)

Let $v \in L^1$ be non-negative, radial and of compact support. Then for $a_p^3 \rho \ll 1$

$$e(\rho) \geq \frac{3}{5}(6\pi^2)^{2/3}\rho^{5/3} + \frac{12\pi}{5}(6\pi)^{2/3} a_p^3 \rho^{8/3}(1 + o(1)).$$

Expected from physics literature with $=$ (Efimov–Amus'ya 1965; Efimov 1966; Ding–Zhang 2019)

- Kinetic energy/energy of free gas.
- Interaction energy.

Spin-polarized Fermi gas

Theorem (L.–Seiringer 2024)

Let $v \geq 0$ be radial and of compact support. Then for $a_p^3 \rho \ll 1$

$$\psi(\beta, \mu) = \psi_0(\beta, \mu) - 12\pi c_{\beta,\mu} a_p^3 \rho^{8/3}[1 + o(1)].$$

Lower bound is uniform in $T = 1/\beta \lesssim \rho^{2/3}$. Upper bound is uniform in $T \sim \rho^{2/3}$.

$T_F \sim \rho^{2/3}$ is the **Fermi temperature**. For higher temperatures expect thermal effects stronger than quantum and so gas should behave like classical gas.

- Pressure of free gas.
- Interaction energy.

$$c_{\beta,\mu} = 2\pi \frac{-\operatorname{Li}_{5/2}(-e^{\beta\mu})}{(-\operatorname{Li}_{3/2}(-e^{\beta\mu}))^{5/3}}, \qquad -\operatorname{Li}_s(-e^x) = \frac{1}{\Gamma(s)} \int_0^\infty \frac{t^{s-1}}{e^{t-x}+1}\,\mathrm{d}t$$

Depends on temperature — spin-$\frac{1}{2}$ didn't.

1st order perturbation theory — Naive $a^3\rho^{8/3}$

Zero temperature:

- Take trial state ψ_F ground state of free gas:

$$\lim_{\substack{N,|\Lambda|\to\infty \\ N/|\Lambda|=\rho}} \frac{\langle\psi_F|H_N|\psi_F\rangle}{|\Lambda|} = \frac{3}{5}(6\pi^2)^{2/3}\rho^{5/3} + \frac{1}{2}\int \rho^{(2)}(x,0)v(x)\,\mathrm{d}x.$$

- Taylor expand $\rho^{(2)}(x,0) \simeq \frac{(6\pi^2)^{2/3}}{5}\rho^{8/3}|x|^2$ and use $\int |x|^2 v(x)\,\mathrm{d}x \approx 24\pi a_p^3$

$\rightsquigarrow$ Energy density $\frac{3}{5}(6\pi^2)^{2/3}\rho^{5/3} + \frac{12\pi}{5}(6\pi^2)^{2/3}a_p^3\rho^{8/3}$.

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Positive temperature:

- Compute pressure functional of free Gibbs state Γ_F:

$$\lim_{L\to\infty} P_\Lambda[\Gamma_F] = \psi_0(\beta,\mu) - \frac{1}{2}\int \rho^{(2)}(x,0)v(x)\,\mathrm{d}x.$$

- Taylor expand $\rho^{(2)}(x,0) \simeq c_{\beta,\mu}\rho^{8/3}|x|^2$ and use $\int |x|^2 v(x) \approx 24\pi a_p^3$.

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$\rightsquigarrow$ Pressure $\psi_0 - 12\pi c_{\beta,\mu}a_p^3\rho^{8/3}$. Explains temperature dependence.

Need to treat interactions more precisely

$\rightsquigarrow$ 2nd order perturbation theory. (Lower bound on $e(\rho)$ and upper bound on $\psi(\beta,\mu)$.)

$\rightsquigarrow$ Cluster expansion. (Upper bound on $e(\rho)$ and lower bound on $\psi(\beta,\mu)$.)

2nd order perturbation theory

$$H_N = H^0 + V, \qquad H^0 = \sum_j -\Delta_j, \qquad V = \sum_{i<j} v_{ij} = \sum_{i<j} v(x_i - x_j).$$

Treat only "off-diagonal" part of V as the perturbation:

$$V = \sum_{i<j} v_{ij} \approx \underbrace{\sum_{i<j} P_i P_j v_{ij} P_i P_j + Q_i Q_j v_{ij} Q_i Q_j}_{V_{\mathrm{D}}:=} + \underbrace{\sum_{i<j} Q_i Q_j v_{ij} P_i P_j + P_i P_j v_{ij} Q_i Q_j}_{V_{\mathrm{OD}}:=}.$$

P and Q projections inside and outside Fermi ball.

Ground state is $\approx e^B \psi_F$ for appropriate unitary e^B.

$$e^{-B} H_N e^B \approx H^0 + V_{\mathrm{D}} + \Big(V_{\mathrm{OD}} + [H^0 + V_{\mathrm{D}}, B] \Big) + \Big([V_{\mathrm{OD}}, B] + \frac{1}{2}[[H^0 + V_{\mathrm{D}}, B], B] \Big).$$

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$$H_N = H^0 + V, \qquad H^0 = \sum_j -\Delta_j, \qquad V = \sum_{i<j} v_{ij} = \sum_{i<j} v(x_i - x_j).$$

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B chosen to cancel off-diagonal part: $[H^0 + V_{\mathrm{D}}, B] \approx -V_{\mathrm{OD}}$. Achieved with (by neglecting 3-body terms and writing $Q \approx \mathbb{1}$)

$$B = \sum_{i<j} P_i P_j (1 - f_p)_{ij} Q_i Q_j - Q_i Q_j (1 - f_p)_{ij} P_i P_j$$

since f_p solves *p-wave scattering equation* (Euler–Lagrange equation of minimization problem)

$$2|x|^2 \Delta f_p + 4x \cdot \nabla f_p - |x|^2 v f_p = 0.$$

2nd order perturbation theory

Conclude

$$\inf_{\|\psi\|=1} \langle \psi | H_N | \psi \rangle \approx \left\langle \psi_F \middle| H^0 \middle| \psi_F \right\rangle + \langle \psi_F | V_{\mathrm{D}} | \psi_F \rangle + \frac{1}{2} \langle \psi_F | [V_{\mathrm{OD}}, B] | \psi_F \rangle \,.$$

with

$$V_{\mathrm{D}} = \sum_{i<j} P_i P_j v_{ij} P_i P_j + Q_i Q_j v_{ij} Q_i Q_j, \qquad V_{\mathrm{OD}} = \sum_{i<j} Q_i Q_j v_{ij} P_i P_j + P_i P_j v_{ij} Q_i Q_j,$$

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2nd order perturbation theory

Conclude

$$\inf_{\|\psi\|=1} \langle\psi|H_N|\psi\rangle \approx \left\langle \psi_F \middle| H^0 \middle| \psi_F \right\rangle + \langle\psi_F|V_{\mathrm{D}}|\psi_F\rangle + \frac{1}{2}\langle\psi_F|[V_{\mathrm{OD}}, B]|\psi_F\rangle\,.$$

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$$B = \sum_{i<j} P_i P_j (1-f_p)_{ij} Q_i Q_j - Q_i Q_j (1-f_p)_{ij} P_i P_j.$$

Compute $[V_{\mathrm{OD}}, B]$ by neglecting 3-body terms and writing $Q \approx \mathbb{1}$:

$$[V_{\mathrm{OD}}, B] \approx -2 \sum_{i<j} P_i P_j v_{ij} (1-f_p)_{ij} P_i P_j$$

Conclude

$$e(\rho) \approx \frac{3}{5}(6\pi^2)^{2/3}\rho^{5/3} + \frac{1}{2}\int \rho^{(2)}(x,0)v(x)\,\mathrm{d}x - \frac{1}{2}\int \rho^{(2)}(x,0)v(x)(1-f_p(x))\,\mathrm{d}x$$

Taylor $\rho^{(2)}(x,0) \simeq \frac{(6\pi^2)^{2/3}}{5}\rho^{8/3}|x|^2$ and use $\int |x|^2 f_p v = 24\pi a_p^3$. Conclude

$$e(\rho) \approx \frac{3}{5}(6\pi^2)^{2/3}\rho^{5/3} + \frac{12\pi}{5}(6\pi^2)^{2/3} a_p^3 \rho^{8/3}.$$

Cluster expansion

Recall from two-particle problem:

The interacting ground state is given by the non-interacting ground state multiplied by the scattering function $f_{s/p}$.

$\rightsquigarrow$ Use f_p to construct better trial state. Jastrow-type trial state:

$$\psi_N(x_1, \ldots, x_N) = \frac{1}{\sqrt{C_N}} \prod_{1 \leq i < j \leq N} f_p(x_i - x_j) D_N(x_1, \ldots, x_N)$$

D_N Slater determinant (with momenta some set M_F, think free ground state ψ_F), C_N normalization constant.

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Compute energy

$$\langle \psi_N | H_N | \psi_N \rangle = E_0 + \iint \rho^{(2)}_{\text{Jas}}(x_1, x_2) \left(\left| \frac{\nabla f_p(x_1 - x_2)}{f_p(x_1 - x_2)} \right|^2 + \frac{1}{2} v(x_1 - x_2) \right) \mathrm{d}x_1 \, \mathrm{d}x_2$$

$$+ \text{3-body term}$$

E_0 kinetic energy of Slater determinant, $\rho^{(2)}_{\text{Jas}}$ the 2-particle density. Think $\rho^{(2)}_{\text{Jas}}(x_1, x_2) \simeq f_p(x_1 - x_2)^2 \rho^{(2)}(x_1, x_2)$. (Recall $12\pi a_p^3 = \int (|\nabla f_p|^2 + \frac{1}{2} v f_p^2)|x|^2 \, \mathrm{d}x$.)

Compute $\rho^{(2)}_{\text{Jas}} \rightsquigarrow$ **Cluster expansion**: Way of calculating $\rho^{(n)}_{\text{Jas}}$. (Formal calculations of Gaudin–Gillespie–Ripka 1971.)

Cluster expansion

$$\rho^{(2)}_{\text{Jas}}(x_1, x_2) = f(x_1 - x_2)^2 \left[\rho^{(2)}(x_1, x_2) + \sum_{p=1}^{\infty} \frac{1}{p!} \sum_{(\pi, G) \in \tilde{\mathcal{L}}^2_p} \Gamma^2_{\pi, G}(x_1, x_2) \right]$$

$$= f(x_1 - x_2)^2 \left[\rho^{(2)}(x_1, x_2) + \int \rho^{(3)} \left(g_{13} + g_{23} + g_{13} g_{23} \right) \mathrm{d}x_3 + \dots \right]$$

Here $g_{ij} = f(x_i - x_j)^2 - 1$. ($\tilde{\mathcal{L}}^2_p$ and $\Gamma^2_{\pi, G}$ are explicit in terms of diagrams.)

Valid for both zero and positive temperature (replacing Slater determinant D_N by free Gibbs state Γ_F at positive temperature)

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$$= f(x_1-x_2)^2 \left[\rho^{(2)}(x_1,x_2) + \int \rho^{(3)} \left(g_{13} + g_{23} + g_{13}g_{23} \right) \mathrm{d}x_3 + \dots \right]$$

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Valid for both zero and positive temperature (replacing Slater determinant D_N by free Gibbs state Γ_F at positive temperature)

Absolute convergence:

$$\sum_{p=1}^{\infty} \frac{1}{p!} \left| \sum_{(\pi,G)\in\tilde{\mathcal{L}}^2_p} \Gamma^2_{\pi,G} \right| < \infty.$$

(Non-trivial) combinatorics $\rightsquigarrow$ Convergence reduces to

$$\rho \int |g(x)| \, \mathrm{d}x \int |\gamma^{(1)}(x;0)| \, \mathrm{d}x = \rho \times a_p^3 \left| \log a_p^3 \rho \right| \times \int_{[0,L]^3} |\gamma^{(1)}(x)| \, \mathrm{d}x < c.$$

$\gamma^{(1)}$ one particle density matrix of Slater determinant D_N. (Or free Gibbs state Γ_F at positive temperature.)

Absolute convergence at zero temperature — The Fermi polyhedron

Absolute convergence

Slater determinant D_N has momenta some set $M_F \subset \frac{2\pi}{L}\mathbb{Z}^3$. This gives $\gamma^{(1)}(x) = \frac{1}{L^3}\sum_{k\in M_F} e^{ikx}$. Hence, convergence criterion reads

$$a_p^3\rho \left|\log a_p^3\rho\right| \times \frac{1}{L^3}\int_{[0,L]^3}\left|\sum_{k\in M_F} e^{ikx}\right| \mathrm{d}x < c.$$

Known as **Lebesgue constant** of a domain. For (Fermi) ball B_F and (Fermi) polyhedron P_F we have (Kolomoitsev–Lomako 2018):

$$\frac{1}{L^3}\int_{[0,L]^3}\left|\sum_{k\in B_F} e^{ikx}\right| \mathrm{d}x \sim N^{1/3}, \qquad \frac{1}{L^3}\int_{[0,L]^3}\left|\sum_{k\in P_F} e^{ikx}\right| \mathrm{d}x \lesssim s(\log N)^3.$$

s is number of corners of P_F – some parameter to choose.

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- Finite size error from kinetic energy $\sim N^{-1/3}\rho^{5/3}$.
- Leading energy density from interaction $\sim a_p^3\rho^{8/3}$.

$\rightsquigarrow$ Need $N^{-1/3}\rho^{5/3} \ll a_p^3\rho^{8/3}$, i.e. $N^{1/3}a_p^3\rho \gg 1$.

$\rightsquigarrow$ **Can't use Fermi ball.**

Absolute convergence at positive temperature

Absolute convergence

$$\gamma^{(1)}(x) = \frac{1}{L^3} \sum_{k \in \frac{2\pi}{L}\mathbb{Z}^3} \frac{1}{1 + e^{\beta|k|^2 - \beta\mu}} e^{ikx}, \qquad \int_\Lambda |\gamma^{(1)}(x)| \, \mathrm{d}x \leq C\rho\beta^{3/2}.$$

Thus, convergence criterion reads

$$a_p^3 \rho \left| \log a_p^3 \rho \right| \beta^{3/2} \rho < c.$$

Uniform in volume. Satisfied for not too small temperatures:

$$T = \frac{1}{\beta} > C\rho^{2/3} (a_p^3 \rho |\log a_p^3 \rho|)^{2/3} \sim T_F (a_p^3 \rho |\log a_p^3 \rho|)^{2/3}.$$

For smaller temperatures we can compare to zero-temperature problem.

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Thank you for your attention!